

Indian Ocean SST and Dipole Outlook

Forecast issued 13 September 2026 · covering 13 September – 12 October 2026

ACCORD / CHC — draft for review

17 September 2026

Abstract

A forecast of sea surface temperature in the two Indian Ocean Dipole boxes, and of the dipole, for the next 30 days and each of the next four weeks, from the ECMWF sub-seasonal (S2S) forecast corrected against NOAA OI SST by linear regression. Third issuance, run on the 13 September initialisation at Chris Funk's request.

The dipole has strengthened again and now exceeds +1.0 over the next 30 days: +0.94 in week 1, +1.13 in week 2, +1.02 across days 1–30 — up from +0.84 on the 7 September run and +0.72 on 31 August. The driver has changed: the western box remains warm (+0.90 to +1.07 above its 2006–2025 average) but the **eastern box has now turned slightly cool** (–0.00 to –0.07, having been +0.08 to +0.16 a week ago). A cooling east is the classic second half of a positive dipole.

Our numbers reproduce Chris's own almost exactly. Comparing like for like — raw model anomaly against the reforecast-mean climatology, which is how his workbook computes it — the western box agrees to **0.005–0.024 °C** at every horizon. The eastern box runs 0.05–0.07 °C warmer in ours, most likely because his climatology uses 2006–2024 and ours 2006–2025; the extra year matters most where warming is fastest.

The eastern gain over persistence is now the durable finding. It clears zero at weeks 2, 3, 4 and on the 30-day window this run, was positive at every horizon in the two previous runs, and cleared at weeks 2 or 4 in both. The west has not cleared zero in any of the three issuances. On the accumulated evidence the product's value is the **eastern box at weeks 2–4**; the west is not distinguishable from persistence.

Method

Model. ECMWF S2S extended-range ensemble, issued 13 September 2026: 100 members, 46 daily steps, 1.5° grid, from the ECMWF Data Store (`ecds.ecmwf.int`), dataset `s2s-forecasts`, variable `sea_surface_temperature`. **Training:** the reforecast set for the same calendar date (`s2s-reforecasts`) — the same run for 13 September in each of 2006–2025, 10 members each. **Observations:** NOAA OI SST v2.1, 0.25° daily, from NOAA PSL. **Boxes:** Saji et al. (1999), west 50–70 °E / 10 °S–10 °N and east 90–110 °E / 10 °S–0, area-weighted by cosine of latitude on each dataset's own grid.

Windows. Days 1–7, 8–14, 15–21, 22–28 and 1–30. S2S SST is filed as a 24-hour mean, so day 1 is the **day of the 00Z issuance**, not the day after: week 1 verifies 13 to 19 September. *Note this differs by one day from the convention in Chris's workbook, which labels lead day 1 as the 14th — it follows `cfgrib's valid_time`, which marks the end of the 24-hour averaging period. Worth settling between us before numbers are compared closely.*

Correction. Per box and window, ordinary least squares of observed on forecast temperature across the twenty reforecast years, applied to this year's forecast. Regression rather than quantile matching, because a straight line continues a warming trend outside the historical range while quantile matching pulls values back inside it.

Uncertainty. An 80% prediction interval on Student's *t*, 18 degrees of freedom, carrying both fitted-line uncertainty and residual scatter — not the naive $1.28 \times \text{residual sd}$, because **this forecast sits 2.3–3.2 standard deviations above the training mean in the west**, where omitting the extrapolation term would understate the interval by 20–28%. It reflects how wrong the method has been on past years, not how much this week's 100 members disagree. **Testing** is leave-one-year-out throughout.

Issuance dates. ECMWF files extended-range reforecasts on **odd calendar days of the month** — verified this week: the 11th, 13th and 15th each have a full 20-year suite, the 12th and 16th have none. **Reproducing:** `iod_pipeline.run("2026-09-13")`.

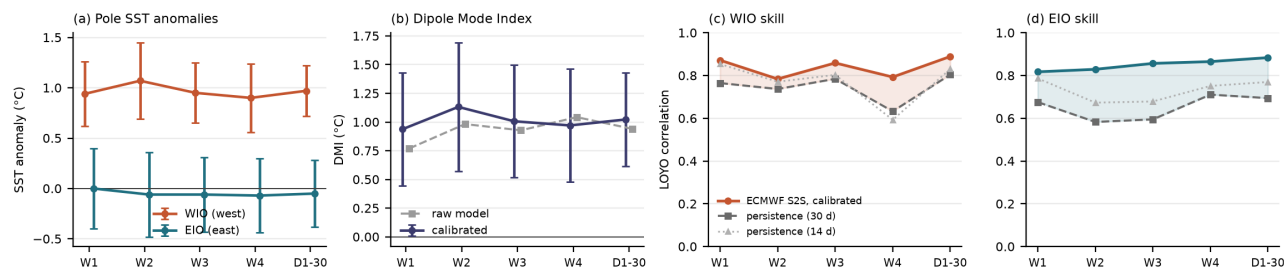


Figure 1: **(a)** Forecast anomaly in each box, 80% prediction intervals. **(b)** The dipole, calibrated and raw — the gap is the regression’s amplitude correction, not bias removal. **(c,d)** Leave-one-year-out correlation against two persistence baselines; shading is the gain over the harder 30-day one.

Results

Horizon	Valid	West box	East box	Dipole
Week 1	13 – 19 Sep	28.72 °C (+0.94)	28.09 °C (–0.00)	+0.94 ± 0.49
Week 2	20 – 26 Sep	28.98 °C (+1.07)	28.12 °C (–0.06)	+1.13 ± 0.56
Week 3	27 Sep – 3 Oct	29.02 °C (+0.95)	28.08 °C (–0.06)	+1.01 ± 0.49
Week 4	4 – 10 Oct	29.19 °C (+0.90)	28.11 °C (–0.07)	+0.97 ± 0.49
Days 1–30	13 Sep – 12 Oct	29.01 °C (+0.97)	28.11 °C (–0.05)	+1.02 ± 0.41

Absolute temperatures; brackets are the departure from the **2006–2025** average for the same dates, dipole in °C with its 80% interval. That baseline is the reforecast period, not the 1991–2020 climatology BoM, CPC and NOAA use — **these anomalies are not directly comparable with the dipole values those centres publish.**

The raw model dipole is +0.77 to +1.04. That difference is **the regression’s amplitude correction, not bias removal**: least squares with an intercept preserves the mean, so constant bias cancels on both sides. What changes is amplitude — the west is stretched (slope 0.95–1.22) and the east shrunk (0.66–0.70). Given the verification record below, that eastern shrinkage remains the part to distrust.

Cross-check against CHC’s own calculation

Chris’s workbook for this same initialisation computes box means independently. Comparing on his indexing, raw model anomaly against the reforecast-mean climatology:

	our raw anomaly	CHC	difference
West, week 1	+0.767	+0.773	–0.006
West, days 1–30	+0.868	+0.880	–0.012
East, week 1	–0.003	–0.050	+0.047
East, days 1–30	–0.072	–0.130	+0.058

The western agreement — within 0.024 °C at every horizon — is an independent check on the box definition, area weighting and lead handling, computed through a different pipeline. The eastern offset of 0.05–0.07 °C is systematic and is most plausibly the climatology period: 2006–2024 in his, 2006–2025 in ours. Worth reconciling, since it propagates directly into the dipole (ours is 0.05–0.09 °C lower than his throughout).

Accuracy

Leave-one-year-out correlation over the twenty reforecast years against persistence — “the ocean stays as it is at issue”. Gains carry an 80% bootstrap interval.

	Week 1	Week 2	Week 3	Week 4	Days 1–30
West — model	0.87	0.78	0.86	0.79	0.89

	Week 1	Week 2	Week 3	Week 4	Days 1–30
persistence 30 d / 14 d	0.76 / 0.85	0.74 / 0.77	0.78 / 0.80	0.63 / 0.59	0.80 / 0.83
gain (80% interval)	+0.11 (–0.03, +0.24)	+0.05 (–0.07, +0.21)	+0.07 (–0.04, +0.17)	+0.16 (–0.01, +0.27)	+0.08 (–0.00, +0.18)
East — model	0.82	0.83	0.86	0.87	0.88
persistence 30 d / 14 d	0.68 / 0.79	0.58 / 0.67	0.59 / 0.68	0.71 / 0.75	0.69 / 0.77
gain (80% interval)	+0.14 (–0.04, +0.27)	+0.25 (+0.00, +0.37)	+0.26 (+0.04, +0.37)	+0.15 (+0.03, +0.22)	+0.19 (+0.02, +0.29)

The east clears zero at four of five horizons this run, the strongest showing in three issuances. Set against the previous two — week 2 in both boxes on 31 August, weeks 4 and days 1–30 in the east on 7 September — the pattern across runs is that **the eastern gain at weeks 2–4 is consistently positive and often significant, while the west has never cleared zero in any run**. Which specific horizons clear still moves between issuances, so read the consistency rather than any single interval.

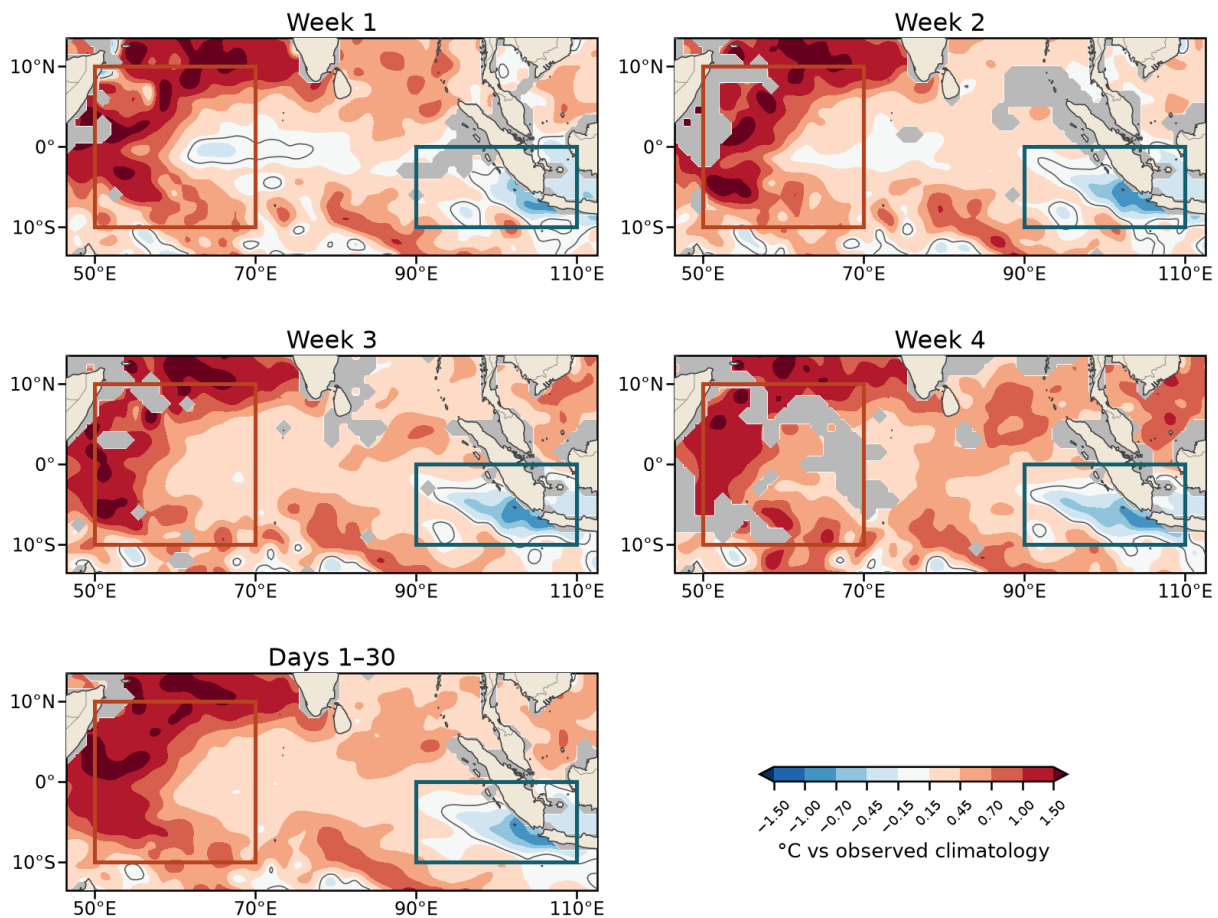


Figure 2: Corrected SST anomaly at every horizon (°C vs the 2006–2025 observed average). Boxes: western (orange) and eastern (teal); land in tan. Grey ocean is where leave-one-year-out correlation falls below 0.4 and the forecast should not be relied on. Each cell is corrected by its own regression, observations averaged from 0.25° to the model’s 1.5° grid. Fields are cubic-refined for contouring — presentational only, and the zero contour in particular is placed by the interpolation rather than measured.

Verification record

Week 1 of this run (13–19 September) is not yet complete. Three windows from earlier issuances have verified:

window	forecast	observed	error	east: raw vs calibrated
31 Aug – 6 Sep	+0.43	+0.01	+0.42	raw exact, calibrated –0.32
7 – 13 Sep (from 31 Aug run)	+0.82	+0.56	+0.26	raw +0.04, calibrated –0.18
7 – 13 Sep (from 7 Sep run)	+0.76	+0.56	+0.20	raw –0.02, calibrated –0.23

Every error is inside its 80% interval, and the shorter lead beat the longer one on the same week. But two patterns now hold across all three: **the dipole forecast runs too positive**, and **the eastern calibration makes the forecast worse** — the raw eastern value has been within 0.04 °C every time, while the calibrated value has been 0.18–0.32 °C too cool. Since the dipole is west minus east, a too-cool east inflates it. Given this run forecasts the strongest dipole yet, that bias is the main reason for caution.

Limitations

- **The eastern calibration has hurt in all three verified windows**, by 0.18–0.32 °C, while the raw eastern value was within 0.04 °C each time. Read the eastern box and the dipole as upper estimates; revisiting the eastern shrinkage is the clearest improvement available.
- **The dipole forecast has run too positive in every verified window** (+0.42, +0.26, +0.20).
- **Which horizons clear significance moves between issuances.** The durable finding across three runs is the eastern gain at weeks 2–4; the west has never cleared.
- **The baseline is 2006–2025, not 1991–2020**, and on a different SST analysis from BoM's, so these anomalies are not directly comparable with published dipole values. Our eastern climatology also differs from CHC's by one year, worth 0.05–0.07 °C.
- **Twenty years is a short training record**, and this forecast extrapolates 2.3–3.2 standard deviations beyond the training mean in the west.
- **The dipole is built from separately calibrated poles**, which amplifies it relative to regressing the observed dipole directly on the model dipole.
- **The interval is historical**, not ensemble-based; a version using the 100 members' spread is the obvious next step.
- **Persistence is given a one-day advantage**, so the comparisons are conservative.
- **Map and box numbers differ by 0.15–0.35 °C**: quote the table for a box.
- **One model only**, ECMWF S2S.